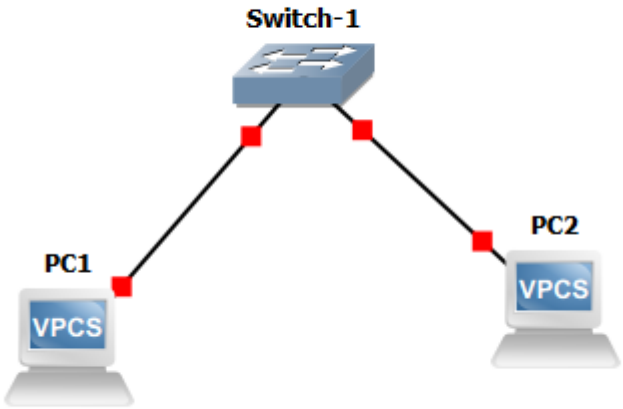
Лабораторная работа №1

Тема: Освоение инструментария для выполнения работ, построение простой сети

2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети



Модель сети 1.

Смена адреса на PC1

PC1> ip 192.168.10.5 255.255.255.0

Checking for duplicate address...

PC1 : 192.168.10.5 255.255.255.0

PC1>

Смена адреса на PC2

PC2> ip 192.168.10.6 255.255.255.0

Checking for duplicate address...

PC2 : 192.168.10.6 255.255.255.0

PC2>

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

Выполнение команды ping с PC2 на PC1.

PC2> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=10.027 ms

84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=0.712 ms

84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=5.841 ms

84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=7.177 ms

84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=5.449 ms

PC2>

4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
20	13.829442	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.6? (ARP Probe)
21	14.829721	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.6? (ARP Probe)
23	15.830535	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.6? (ARP Probe)
59	68.453454	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request)
61	69.453883	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request)
63	70.453949	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request)
68	77.848774	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request) (duplicate use of 192.168.10.5 detected!)
69	77.848886	Private_66:68:00	Private_66:68:01	ARP	64	Gratuitous ARP for 192.168.10.5 (Reply)
75	85.565817	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.6 (Request)
76	86.566337	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.6 (Request)
78	87.566642	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.6 (Request)
86	95.533425	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.6? Tell 192.168.10.5
87	95.534247	Private_66:68:01	Private_66:68:00	ARP	64	192.168.10.6 is at 00:50:79:66:68:01

Пакеты, перехваченные на узле PC1

No.	Time	Source	Destination	Protocol	Lengtl	Info
56	68.454057	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request)
58	69.454351	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request)
60	70.454388	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request)
65	77.848625	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.5 (Request) (duplicate use of 192.168.10.5 detected!)
66	77.849154	Private_66:68:00	Private_66:68:01	ARP	64	Gratuitous ARP for 192.168.10.5 (Reply)
72	85.565623	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.6 (Request)
73	86.566138	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.6 (Request)
75	87.566520	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.10.6 (Request)
83	95.533949	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.6? Tell 192.168.10.5
84	95.534175	Private_66:68:01	Private_66:68:00	ARP	64	192.168.10.6 is at 00:50:79:66:68:01

Пакеты, перехваченные на узле PC2

Пакет-запрос 86 на PC1 – это broadcast запрос от PC1 выполненный по команде ping 192.168.10.6 чтобы узнать у какого компьютера такой IP-адрес. Пакет-ответ 87 на PC1 – это unicast ответ от PC2 на запрос PC1 с указанием MAC-адреса PC2.

Пакет 86 (запрос)

Размером 64 байта, внутри инкапсулирован протокол Ethernet

```
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast
(ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
        .... ..1. .... .... .... .... = LG bit: Locally administered address
(this is NOT the factory default)
        .... ...1 .... .... .... .... = IG bit: Group address
(multicast/broadcast)
    Source: Private_66:68:00 (00:50:79:66:68:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 5]
    Padding: 000000000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
```

Из полей Destination и Source видно, что используется широковещательный адрес (ff:ff:ff:ff:ff:ff) для конечного адреса, а в качестве источника указан (00:50:79:66:68:00).

```
Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
    Sender IP address: 192.168.10.5
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.10.6
```

Из текста выше, видно, что используется протокол IPv4, MAC и IP адреса отправителя и конечного адресов.

Пакет 87 (ответ)

Размером также 64 байт и инкапсулирован Ethernet.

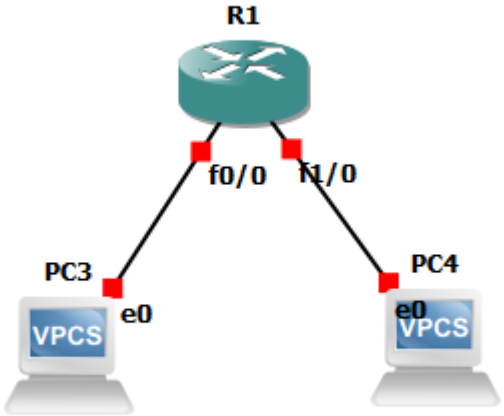
```
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00
(00:50:79:66:68:00)
    Destination: Private_66:68:00 (00:50:79:66:68:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: Private_66:68:01 (00:50:79:66:68:01)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 7]
    Padding: 000000000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
```

Видно, что адреса назначения уже не широковещательный, а уни-каст, как и адрес отправителя.

```
Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
    Sender IP address: 192.168.10.6
    Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
    Target IP address: 192.168.10.5
```

Тут уже видно, что целевой MAC-адрес не широковещательный, а конкретного устройства.

5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей



Модель сети 2

На маршрутизаторе

Включаем привилегированный режим, входим в терминал, настраиваем каждый из портов: адрес и маска, включаем интрефейс на порту, а в конце включаем маршрутизацию, чтоб самому не писать ничего в таблицу.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fastEhernet
R1(config)#interface fastEhernet 0/0
                                ^
% Invalid input detected at '^' marker.

R1(config)#interf
R1(config)#interface fastE
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:01:53.679: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar  1 00:01:54.679: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R1(config)#interface fastEthernet 1/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:02:23.123: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar  1 00:02:24.123: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R1(config)#ip routing
R1(config)#end
R1#
*Mar  1 00:02:33.655: %SYS-5-CONFIG_I: Configured from console by console
```

```
R1#
```

На PC3

Настройка адреса, шлюза и маски. В качестве шлюза указывается адрес соответствующего порта маршрутизатора.

```
PC3> ip 192.168.1.10 192.168.1.1 24
Checking for duplicate address...
PC3 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC3> save
Saving startup configuration to startup.vpc
. done
```

На PC4

```
PC4> ip 192.168.2.20 192.168.2.1 24
Checking for duplicate address...
PC4 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC4> save
Saving startup configuration to startup.vpc
.  done
```

Проверка настройки маршрутизатора и PC

На маршрутизаторе

```
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status
Prot
ocol
FastEthernet0/0            192.168.1.1     YES manual up
up
FastEthernet1/0            192.168.2.1     YES manual up
up
Ethernet2/0                unassigned      YES unset  administratively down
down
Ethernet2/1                unassigned      YES unset  administratively down
down
Ethernet2/2                unassigned      YES unset  administratively down
down
Ethernet2/3                unassigned      YES unset  administratively down
down
Serial3/0                  unassigned      YES unset  administratively down
down
Serial3/1                  unassigned      YES unset  administratively down
down
Serial3/2                  unassigned      YES unset  administratively down
down
Serial3/3                  unassigned      YES unset  administratively down
down
```

На ПК3

```
PC3> show ip

NAME        : PC3[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:03
LPORT       : 20036
RHOST:PORT  : 127.0.0.1:20037
MTU         : 1500
```

На ПК4

```
PC4> show ip

NAME        : PC4[1]
IP/MASK     : 192.168.2.20/24
GATEWAY     : 192.168.2.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT       : 20038
RHOST:PORT  : 127.0.0.1:20039
MTU         : 1500
```

Проверка шлюзов

PC3> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=9.457 ms

84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=6.542 ms

84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=6.793 ms

84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=6.582 ms

84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=5.769 ms

PC4> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=10.047 ms

84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=5.693 ms

84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=7.036 ms

84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=6.690 ms

84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=6.001 ms

6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

На ПК3

PC3> ping 192.168.2.20

84 bytes from 192.168.2.20 icmp_seq=1 ttl=63 time=19.106 ms

84 bytes from 192.168.2.20 icmp_seq=2 ttl=63 time=16.790 ms

84 bytes from 192.168.2.20 icmp_seq=3 ttl=63 time=16.116 ms

84 bytes from 192.168.2.20 icmp_seq=4 ttl=63 time=15.720 ms

84 bytes from 192.168.2.20 icmp_seq=5 ttl=63 time=16.496 ms

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

2	0.010050	cc:01:0f:fd:00:00	Broadcast	ARP	60	Gratuitous ARP for 192.168.1.1 (Reply)
3	0.010075	cc:01:0f:fd:00:00	Broadcast	ARP	60	Gratuitous ARP for 192.168.1.1 (Reply)
46	352.037887	Private_66:68:03	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.10 (Request)
47	353.038686	Private_66:68:03	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.10 (Request)
48	354.039530	Private_66:68:03	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.10 (Request)
116	915.537910	Private_66:68:03	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
117	915.541550	cc:01:0f:fd:00:00	Private_66:68:03	ARP	60	192.168.1.1 is at cc:01:0f:fd:00:00
118	915.542242	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xd33c, seq=1/256, ttl=64 (reply in 119)
119	915.551627	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xd33c, seq=1/256, ttl=255 (request in 118)
120	916.551941	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xd43c, seq=2/512, ttl=64 (reply in 121)
121	916.558377	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xd43c, seq=2/512, ttl=255 (request in 120)
122	917.559619	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xd53c, seq=3/768, ttl=64 (reply in 123)
123	917.566302	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xd53c, seq=3/768, ttl=255 (request in 122)
124	918.567184	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xd63c, seq=4/1024, ttl=64 (reply in 125)
125	918.573647	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xd63c, seq=4/1024, ttl=255 (request in 124)
127	919.574712	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xd73c, seq=5/1280, ttl=64 (reply in 128)
128	919.580390	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xd73c, seq=5/1280, ttl=255 (request in 127)
136	985.233881	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x193d, seq=1/256, ttl=64 (reply in 137)
137	985.252886	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x193d, seq=1/256, ttl=63 (request in 136)
138	986.253087	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1a3d, seq=2/512, ttl=64 (reply in 139)
139	986.269751	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1a3d, seq=2/512, ttl=63 (request in 138)
140	987.270217	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1b3d, seq=3/768, ttl=64 (reply in 141)
141	987.286226	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1b3d, seq=3/768, ttl=63 (request in 140)
142	988.287371	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1c3d, seq=4/1024, ttl=64 (reply in 143)
143	988.302987	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1c3d, seq=4/1024, ttl=63 (request in 142)
145	989.303371	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1d3d, seq=5/1280, ttl=64 (reply in 146)
146	989.319751	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1d3d, seq=5/1280, ttl=63 (request in 145)

Трафик на PC3

2	0.000053	cc:01:0f:fd:00:10	Broadcast	ARP	60	Gratuitous ARP for 192.168.2.1 (Reply)
3	0.000059	cc:01:0f:fd:00:10	Broadcast	ARP	60	Gratuitous ARP for 192.168.2.1 (Reply)
78	611.687141	Private_66:68:02	Broadcast	ARP	64	Gratuitous ARP for 192.168.2.20 (Request)
79	612.687656	Private_66:68:02	Broadcast	ARP	64	Gratuitous ARP for 192.168.2.20 (Request)
80	613.688674	Private_66:68:02	Broadcast	ARP	64	Gratuitous ARP for 192.168.2.20 (Request)
113	894.539141	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.2.1? Tell 192.168.2.20
114	894.541166	cc:01:0f:fd:00:10	Private_66:68:02	ARP	60	192.168.2.1 is at cc:01:0f:fd:00:10
115	894.541253	192.168.2.20	192.168.2.1	ICMP	98	Echo (ping) request id=0xdb3c, seq=1/256, ttl=64 (reply in 116)
116	894.551237	192.168.2.1	192.168.2.20	ICMP	98	Echo (ping) reply id=0xdb3c, seq=1/256, ttl=255 (request in 115)
117	895.552006	192.168.2.20	192.168.2.1	ICMP	98	Echo (ping) request id=0xdc3c, seq=2/512, ttl=64 (reply in 118)
118	895.557587	192.168.2.1	192.168.2.20	ICMP	98	Echo (ping) reply id=0xdc3c, seq=2/512, ttl=255 (request in 117)
119	896.558442	192.168.2.20	192.168.2.1	ICMP	98	Echo (ping) request id=0xdd3c, seq=3/768, ttl=64 (reply in 120)
120	896.565364	192.168.2.1	192.168.2.20	ICMP	98	Echo (ping) reply id=0xdd3c, seq=3/768, ttl=255 (request in 119)
121	897.565931	192.168.2.20	192.168.2.1	ICMP	98	Echo (ping) request id=0xde3c, seq=4/1024, ttl=64 (reply in 122)
122	897.572522	192.168.2.1	192.168.2.20	ICMP	98	Echo (ping) reply id=0xde3c, seq=4/1024, ttl=255 (request in 121)
123	898.573473	192.168.2.20	192.168.2.1	ICMP	98	Echo (ping) request id=0xdf3c, seq=5/1280, ttl=64 (reply in 124)
124	898.579345	192.168.2.1	192.168.2.20	ICMP	98	Echo (ping) reply id=0xdf3c, seq=5/1280, ttl=255 (request in 123)
132	955.804071	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x193d, seq=1/256, ttl=63 (reply in 133)
133	955.804210	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x193d, seq=1/256, ttl=64 (request in 132)
134	956.820913	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1a3d, seq=2/512, ttl=63 (reply in 135)
135	956.821047	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1a3d, seq=2/512, ttl=64 (request in 134)
136	957.837418	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1b3d, seq=3/768, ttl=63 (reply in 137)
137	957.837556	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1b3d, seq=3/768, ttl=64 (request in 136)
138	958.854172	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1c3d, seq=4/1024, ttl=63 (reply in 139)
139	958.854273	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1c3d, seq=4/1024, ttl=64 (request in 138)
141	959.870943	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x1d3d, seq=5/1280, ttl=63 (reply in 142)
142	959.871072	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1d3d, seq=5/1280, ttl=64 (request in 141)

Трафик на PC4

В отличии от ARP-пакетов, которые были по 60-64 байта (в зависимости от того, была ли проверена контрольная сумма или нет) пакеты ICMP имеют размер 98 байт.

Анализ пакета 4 (запрос на выдачу IP адреса)

```
Ethernet II, Src: cc:01:0f:fd:00:00 (cc:01:0f:fd:00:00), Dst: Broadcast
(ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
        .... ..1. .... .... .... .... = LG bit: Locally administered address
(this is NOT the factory default)
        .... ...1 .... .... .... .... = IG bit: Group address
(multicast/broadcast)
    Source: cc:01:0f:fd:00:00 (cc:01:0f:fd:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 1]
    Padding: 000000000000000000000000000000000000
```

В качестве MAC-адреса отправителя тут указан MAC порта. Адрес назначения широковещательный.

```
Address Resolution Protocol (reply/gratuitous ARP)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    [Is gratuitous: True]
    Sender MAC address: cc:01:0f:fd:00:00 (cc:01:0f:fd:00:00)
    Sender IP address: 192.168.1.1
```

```
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.1.1
```

Тут у отправителя и получателя совпадают IP-адреса, но MAC – разные, у отправителя свой собственный, а у получателя широковещательный.

Анализ пакета 136 (ping запрос)

```
Ethernet II, Src: Private_66:68:03 (00:50:79:66:68:03), Dst:
cc:01:0f:fd:00:00 (cc:01:0f:fd:00:00)
    Destination: cc:01:0f:fd:00:00 (cc:01:0f:fd:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: Private_66:68:03 (00:50:79:66:68:03)
        .... ..0. .... .... .... .... = LG bit: Globally unique address
(factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 5]
Запрос идёт на конкретный MAC-адрес, но тут запрос идёт на MAC-адрес шлюза, а
не MAC-адрес второго компьютера (у которого MAC = Private_66:68:02).
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.2.20
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x3d19 (15641)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0xb921 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.10
    Destination Address: 192.168.2.20
    [Stream index: 1]
```

Тут адрес назначения и адрес источника совпадают с теми, что установлены на PC4 и PC3

```
Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x06ce [correct]
    [Checksum Status: Good]
    Identifier (BE): 6461 (0x193d)
    Identifier (LE): 15641 (0x3d19)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 137]
    Data (56 bytes)
```

Контрольная сумма совпала.

```
        Data:
08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2
e2f303132333435363738393a3b3c3d3e3f
```

```
        [Length: 56]
```

Так же имеется поле с данными, которое заполнено определённой последовательностью байтов.

Если проанализировать аналогичные пакеты, но на PC4, то там будет замены MAC-адреса на такие как у PC4 и шлюза на который он посылает ответ.